

MUGDHA TADWALKAR

Bengaluru, Karnataka, India

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Summary

Agentic AI Developer with 1+ years designing and deploying multi-agent systems, LLM-driven pipelines, and autonomous AI solutions in enterprise environments. Hands-on with Python, LangChain, CrewAI, RAG, Agentic Orchestration, and Vector Databases – building agents capable of reasoning, planning, and executing tasks end-to-end. Proficient in LLM Integration & Deployment, FastAPI, REST APIs, CI/CD, and cloud platforms (Azure/AWS) with full-lifecycle ownership from POC to production deployment.

Education

JSPM's College of Engineering	2020 – 2024
B.E in Computer Science and Engineering	Pune, India

Experience

Software Engineer – AI & Data Solutions	May 2025 – Present
Tata Consultancy Services (TCS)	Bengaluru, India
- Developed AI-powered document extraction workflows for a UK pension administration platform using Python, automating the processing of pension documents including SOD, LOA, and related forms. - Engineered multimodal LLM pipelines using Gemma 3/Gemini with OCR integration for document classification and structured data extraction from scanned PDFs and email attachments. - Designed advanced prompt engineering strategies and JSON schema-driven extraction pipelines to accurately capture member, deceased, notifier, and death certificate information. - Implemented response validation, guardrails, and error-handling mechanisms to improve extraction consistency and reduce incorrect document classifications in production workflows. - Integrated AI services with enterprise APIs and backend components, contributing to scalable document-processing workflows for UK pension operations. - Collaborated with cross-functional teams in an Agile/Scrum environment to deliver production-ready AI features, perform UAT support, and continuously improve model performance through iterative experimentation.	

Projects

LeadCommand – Scalable Multi-Agent AI Platform	2025 – Present
Python, FastAPI, CrewAI, LangChain, Groq/LLaMA, Supabase, Next.js, GitHub Actions	
- Architected a production multi-agent AI platform using CrewAI – designing autonomous agents with tool-use, memory, and agentic orchestration for real estate ad automation end-to-end. - Built modular, independently deployable backend services with FastAPI and CI/CD (GitHub Actions) – applying best practices in guardrails, error handling, observability, and REST API design.	
RAG-Powered IT Helpdesk – Agentic NLP System	2025
Python, FastAPI, ChromaDB, Groq/LLaMA, LangChain, TF-IDF, Streamlit	
Built an autonomous agent combining semantic RAG retrieval (ChromaDB vector database) with an NLP classifier – implemented fallback mechanisms and evaluation pipelines to validate reliability before enterprise rollout.	

Technical Skills

Programming: Python, SQL, Java, Pandas, NumPy
Agentic AI & Orchestration: LangChain, CrewAI, LangGraph (fundamentals), AutoGen (fundamentals), Multi-Agent Systems, Agentic Orchestration, Prompt Engineering
LLMs & Integration: LLM Integration & Deployment, RAG, LLaMA, GPT-4o, DeepSeek
Vector Databases & Memory: ChromaDB, FAISS (fundamentals), Pinecone (fundamentals), Semantic Search, Embedding Pipelines
Backend & Deployment: FastAPI, REST APIs, Microservices, CI/CD (GitHub Actions), Docker (fundamentals), Azure, AWS
ML & NLP: Scikit-learn, XGBoost, NLP, Feature Engineering, Model Evaluation, Statistical Analysis
Databases: PostgreSQL, MySQL, MongoDB, Supabase, ChromaDB

Certifications

Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate
Problem Solving (Intermediate) – HackerRank